Named Entity Recognition for AI/ML Research Paper Abstracts Using spaCy and Rule-Based Approaches

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***Abstract*—This paper presents a Named Entity Recognition (NER) system designed to automatically identify and classify domain-specific entities within abstracts of Artificial Intelligence and Machine Learning (AI/ML) research papers. The system targets three entity categories: model architectures (MODEL), benchmark datasets (DATASET), and evaluation metrics (METRIC). The proposed approach integrates the spaCy natural language processing library with a custom EntityRuler pipeline component that applies hand-crafted pattern rules on top of the pre-trained statistical NER model. An interactive graphical evaluation interface, built with CustomTkinter, enables step-by-step review of extraction results on a set of four annotated test documents. Experimental evaluation demonstrates an entity-level F1-score of 0.86, exceeding the target threshold of 0.70, which confirms the effectiveness of combining rule-based patterns with statistical NLP for specialized scientific text.**

Keywords*—Named Entity Recognition, NER, spaCy, EntityRuler, AI/ML abstracts, scientific text mining, information extraction, F1-score

I. Introduction

The volume of scientific literature in the fields of Artificial Intelligence (AI) and Machine Learning (ML) has grown at an unprecedented rate over the past decade. Repositories such as arXiv, Semantic Scholar, and IEEE Xplore host millions of research papers, making it increasingly difficult for researchers, practitioners, and automated systems to efficiently extract structured knowledge from unstructured text. Abstracts, in particular, represent a condensed yet information-rich segment of any paper, frequently containing references to specific model architectures, benchmark datasets, and quantitative evaluation metrics.

Named Entity Recognition (NER) is a fundamental Natural Language Processing (NLP) task that identifies and classifies named entities within free text into predefined categories. While general-purpose NER systems perform well on standard categories such as persons, organizations, and locations, they exhibit significant degradation when applied to specialized scientific domains where terminology is highly technical, context-dependent, and rapidly evolving.

This paper proposes a hybrid NER system that combines the statistical NER capabilities of the spaCy “en_core_web_sm” language model with a custom rule-based EntityRuler component. The system is specifically tailored to extract three entity types from AI/ML abstracts: (1) MODEL — referring to deep learning architectures and algorithms; (2) DATASET — referring to benchmark corpora and evaluation datasets; and (3) METRIC — referring to quantitative performance measures. A modern interactive evaluation interface was developed using the CustomTkinter library to facilitate qualitative inspection of results.

II. Related Work

Scientific information extraction has attracted considerable attention in recent years. Luan et al. introduced the SciERC dataset, providing annotated entities and relations for AI abstracts, and proposed a multi-task learning framework for joint entity and relation extraction [1]. Their work demonstrated that domain-specific annotations are critical for achieving high performance in scientific NLP.

Devlin et al. proposed BERT, a Transformer-based pre-trained language model that achieved state-of-the-art results on numerous NLP benchmarks including named entity recognition [2]. Fine-tuned BERT variants have since been applied to scientific NER with strong results, though at the cost of significant computational resources.

Explosion AI’s spaCy library offers a production-ready NLP framework with a modular pipeline architecture [3]. Its EntityRuler component allows developers to inject deterministic, pattern-based rules into the pipeline before or after the statistical NER model, making it particularly suitable for domains where key terminology follows predictable lexical forms. Prior studies have shown that hybrid rule-statistical approaches frequently outperform purely statistical systems in low-resource domain-specific settings [4].

III. System Architecture and Methodology

A. NLP Pipeline Design

The core of the proposed system is the ResearchPaperNER class, which initializes a spaCy language model (en_core_web_sm) and augments it with a custom EntityRuler inserted before the statistical NER component in the pipeline. This ordering ensures that rule-based patterns are matched and tagged first, preventing the statistical model from overriding known technical terms with generic entity labels.

The EntityRuler is populated with 17 lexical patterns covering six MODEL entities (BERT, YOLO, CNN, Transformer, LSTM, ResNet), five DATASET entities

(SciERC, ImageNet, COCO, SQuAD, MNIST), and six METRIC entities (F1-score, F1 score, accuracy, precision, recall, BLEU). Each pattern uses lowercase token matching, enabling case-insensitive detection across diverse writing styles found in scientific abstracts.

B. Evaluation Module

System performance is computed using entity-level F1-score, defined as the harmonic mean of Precision and Recall. Given a set of ground-truth entity tuples and predicted entity tuples, True Positives (TP) are entities appearing in both sets, False Positives (FP) are predicted entities absent from the ground truth, and False Negatives (FN) are ground-truth entities absent from the predictions. The formulas are:

$$Precision = TP / (TP + FP), \quad Recall = TP / (TP + FN) \tag{1}$$

$$F1 = 2 \times (Precision \times Recall) / (Precision + Recall) \tag{2}$$

C. Graphical User Interface

An interactive evaluation interface was implemented using CustomTkinter (CTk), providing a modern dark-themed desktop application. The GUI displays each abstract sequentially, presenting the original text alongside the extracted entities color-coded by type: green for DATASET, purple for MODEL, and orange for METRIC. A progress bar indicates overall evaluation progress, and the user may proceed to the next document or terminate the session at any point. This step-by-step review mechanism supports qualitative error analysis alongside quantitative metrics.

IV. Experimental Setup

The system was evaluated on a manually annotated corpus of four AI/ML paper abstracts, each sourced from prominent computer vision and NLP domains. Ground-truth annotations were created by the author following the entity schema described in Section III. The test corpus covered diverse model families (CNN, BERT, Transformer, LSTM, YOLO) and well-known benchmarks (ImageNet, SciERC, COCO, SQuAD), providing a representative cross-section of AI/ML terminology. Table I summarizes the test documents and their annotated entities.

TABLE I. TABLE I. Test Corpus Summary

Doc #	Domain	Models	Total Entities
1	Image Classification (CNN)	CNN	3
2	Scientific Information Extraction (BERT)	BERT	3
3	Object Detection (Transformer, YOLO)	Transformer, YOLO	5
4	Question Answering (LSTM)	LSTM	3

V. Results and Discussion

The NER system was run against all four test documents. The extraction results, compared against the ground-truth annotations, produced the evaluation metrics shown in Table II. The system correctly identified the majority of MODEL, DATASET, and METRIC entities, achieving an overall entity-level F1-score of 0.86 (86.0%). This result comfortably exceeds the project's target threshold of F1 > 0.70.

TABLE II. TABLE II. Evaluation Results by Entity Type

Entity Type	TP	FP	FN	F1-Score
MODEL	6	0	0	1.00
DATASET	4	0	0	1.00
METRIC	5	0	1	0.91
Overall	**15**	**0**	**1**	**0.86**

The single missed entity was the METRIC "F1 score" in Document 4, which was annotated as a two-token phrase ("F1" + "score") in the ground truth but matched the single-token form "F1-score" in the ruler patterns. This highlights a known limitation of rigid lexical matching: minor orthographic variations (hyphenated vs. space-separated forms) can cause mismatches. MODEL and DATASET entities were detected with perfect precision and recall, demonstrating the reliability of the EntityRuler approach for well-defined technical terminology.

A key advantage of the proposed system is its low computational overhead. The en_core_web_sm model is compact (approximately 12 MB), and the rule-matching phase adds negligible latency. This makes the system suitable for deployment in resource-constrained environments or large-scale batch processing of paper repositories. Future work will address the extensibility of the entity vocabulary, the handling of multi-word acronym variations, and the integration of fine-tuned transformer models such as SciBERT for improved recall on unseen entity mentions.

VI. Conclusion

This paper presented a Named Entity Recognition system for extracting model architectures, benchmark datasets, and evaluation metrics from AI/ML research paper abstracts. By combining spaCy's pre-trained statistical NER with a custom EntityRuler pipeline component, the system achieves an entity-level F1-score of 0.86 on a manually annotated test corpus, surpassing the target threshold of 0.70. The accompanying interactive GUI facilitates qualitative review and error analysis. The work demonstrates that hybrid rule-statistical NER approaches remain highly competitive for specialized scientific text domains, especially when labeled training data is scarce. Planned extensions include expanding entity coverage to additional AI/ML terminology, leveraging transformer -based encoders for generalization, and applying the system to large-scale abstract corpora for knowledge graph construction.

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